

DATA SCIENCE

COVID-19 / Public Health Data Visualization Dashboard

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ABSTRACT

The COVID-19 / Public Health Data Visualization Dashboard is a Data Science project designed to explore, analyze, and present public health trends through meaningful visualizations. The project focuses on transforming COVID-19 or related public health datasets into clear and understandable information that can help users identify patterns in cases, recoveries, deaths, and other available indicators. The main purpose is to provide a simple interactive platform through which users can examine public health data without having to work directly with large and complex raw datasets.

The project starts with the collection of a suitable public health dataset containing relevant COVID-19 information. The data is then inspected and preprocessed to improve its quality and make it suitable for analysis. Common preparation activities include identifying missing or inconsistent values, organizing the data into an appropriate structure, and selecting relevant attributes for analysis. After preprocessing, Exploratory Data Analysis (EDA) is performed to understand the distribution of the data, identify trends, compare categories or regions, and discover relationships between important variables.

Data visualization forms the central part of the proposed system. Python visualization libraries such as Matplotlib and Seaborn are used to create informative charts and graphs. Depending on the available dataset, the dashboard can present time-based trends, regional comparisons, case distributions, recovery and mortality patterns, and other useful indicators. Different visual representations are selected according to the type of information being analyzed so that complex numerical data can be communicated in a clear and accessible manner.

An interactive dashboard is developed to make the analysis more flexible and user-friendly. Interactive filters allow users to select relevant categories, regions, or time periods and view the corresponding results. Instead of displaying only static graphs, the dashboard provides an organized environment in which users can explore different portions of the dataset and observe how the selected conditions affect the displayed information. This interactive approach makes the system useful for academic analysis, demonstrations, and exploratory public health data studies.

The expected outcome of the project is a functional visualization dashboard that combines data preprocessing, exploratory analysis, statistical visualization, and interactive filtering in one system. The dashboard is intended to make public health trends easier to interpret and to demonstrate how Data Science techniques can be applied to a real-world dataset. It can help users recognize significant changes and patterns in the available data and support data-driven understanding of the pandemic and its public health impact.

Overall, the project demonstrates the practical use of Python-based Data Science techniques, particularly data cleaning, Exploratory Data Analysis, visualization, and dashboard development. By converting raw public health information into interactive visual insights, the system provides a structured and accessible way to

explore COVID-19-related trends while also demonstrating the importance of visual analytics in understanding large datasets.

OBJECTIVES

- Analyze COVID-19/public health data and identify meaningful trends.
- Preprocess the dataset by handling missing or inconsistent data.
- Perform Exploratory Data Analysis to understand important patterns and relationships.
- Create clear visualizations using Matplotlib and Seaborn.
- Develop interactive filters for exploring different parts of the dataset.
- Present complex public health information in a simple and understandable form.

TECHNOLOGIES / TOOLS

Python • Pandas • Matplotlib • Seaborn • Interactive Dashboard Framework • CSV/Public Health Dataset

KEYWORDS

COVID-19, Public Health, Data Science, EDA, Data Preprocessing, Data Visualization, Matplotlib, Seaborn, Interactive Dashboard, Trend Analysis.